

AEQUITAS

Decentralized Human Currency

*"Money exists because people exist.
Nothing more, nothing less."*

Version	v1.0
Date	June 2026
Chain	Aequitas Chain
Chain ID	1926 (0x786)
Status	Phase 0 — Live
Explorer	aequitas-production-9fba.up.railway.app

Contract Addresses

Contract	Address	Network
AequitasV6 (Main)	0xA76cA3bf34F2Ae5dFA0608696627e42b81180488	Aequitas Chain (1926)
BioVerifier (Groth16)	0xc369D27b49DE017d113Bbcb9A1884a9e745B6BE2	Aequitas Chain (1926)
AequitasV5 (Legacy)	0x4f147d5B3388AF07993CC4fC548502A78Af0B8b5	Ethereum Sepolia

Abstract

Aequitas introduces a novel monetary system where the total money supply is mathematically and permanently tied to the number of verified human beings. Every unique human who proves their existence through biometric Zero-Knowledge Proof receives exactly 1,000 AEQ — the first person, the millionth person, and the billionth person all receive identical amounts. Supply can only grow when new humans join; it cannot be created through any other mechanism.

The system runs on a custom Layer 1 blockchain (Aequitas Chain, Chain ID 1926) built in Go, featuring BlockDAG consensus, EVM compatibility, and persistent PostgreSQL state shared across nodes. The V6 economic model introduces Proof of Alive, Guardian System, Demurrage, Wealth Cap, and protocol-native UBI — all enforced by immutable smart contract code without human intervention.

Bitcoin's estimated Gini coefficient exceeds 0.85, higher than any country on Earth. Aequitas was designed to solve this fundamental problem: a monetary system where equality is not a goal but a mathematical guarantee.

1. The Problem with Existing Monetary Systems

1.1 Bitcoin's Inequality Problem

Bitcoin was introduced in 2009 as a peer-to-peer electronic cash system that would democratize finance. The reality has been the opposite. Early miners accumulated millions of coins at near-zero cost. By 2021, the top 1% of Bitcoin addresses controlled over 90% of all Bitcoin in existence. Bitcoin's estimated Gini coefficient exceeds 0.85 — higher than any country on Earth, higher than historical feudal societies, approaching the theoretical maximum of inequality.

1.2 The Pattern Repeats

Ethereum, Solana, and virtually every other cryptocurrency follow the same pattern: pre-mine for founders and early investors, ICO for wealthy participants, and the rest of humanity priced out before they even hear about the project. These systems replicate and amplify the wealth concentration of traditional finance rather than solving it.

1.3 The Root Cause

The fundamental problem is that existing cryptocurrencies treat money as a commodity to be mined, purchased, or earned — rather than as a social institution that exists because humans exist and trade with each other. A monetary system designed for humans should distribute value according to human existence, not according to computational power, capital, or timing.

2. The Aequitas Solution

"Money exists because people exist. Therefore, every person should have an equal share of money simply by virtue of being human."

2.1 Core Formula

$$\text{Total AEQ Supply} = \text{Verified Active Humans} \times 1,000$$

This formula is not a policy target — it is a mathematical law enforced by immutable smart contract code. No person, organization, government, or founding team can violate it without a community-consensus hard fork.

2.2 Proof of Humanity

Every AEQ holder must prove they are a unique living human through biometric verification. This is not optional. Without Proof of Humanity, no AEQ can be received. This eliminates bots, duplicate accounts, corporations, governments, and AI systems from the monetary system — only real humans can participate.

Technical implementation: The user's fingerprint is processed by the Hardware Secure Element (HSE) of their Android smartphone — the same dedicated cryptographic chip that secures banking apps and the phone's own lock screen. The HSE generates a cryptographic signature. This signature is hashed into a field element of the BN128 elliptic curve. A Groth16 Zero-Knowledge Proof is generated from this hash on the Proof Server. The proof guarantees 'a unique biometric hash was used' without revealing anything about the hash itself. The proof is then verified on-chain before registration is permitted.

3. Technical Architecture

3.1 Aequitas Chain

Parameter	Value
Language	Go 1.24
Consensus	BlockDAG + Proof of Humanity
Chain ID	1926 (0x786)
Block Time	~6 seconds average
EVM	Yes — go-ethereum engine (Shanghai)
P2P Protocol	libp2p (Go implementation)
State Storage	PostgreSQL (persistent, shared)
ZKP System	Groth16 / snarkjs / circom
ZKP Curve	BN128 (alt-bn128)
Proof Size	~200 bytes
Verification Time	~10ms
Initial Grant	1,000 AEQ per verified human
Transaction Fee	0.1%
Gas	Gasless for registration (protocol-covered)

3.2 BlockDAG Consensus

Unlike traditional linear blockchains, Aequitas uses a Directed Acyclic Graph (DAG) where multiple blocks can be produced simultaneously by different nodes. When two nodes produce blocks at the same time, both are valid and are later merged into a single 'merge block' with multiple parents. This architecture enables significantly higher throughput, lower latency, and better fault tolerance than single-chain designs.

3.3 Zero-Knowledge Proof System

Parameter	Value
Proving System	Groth16
Elliptic Curve	BN128 (alt-bn128)
Trusted Setup	Hermes Ceremony Parameters

Circuit Compiler	circom + snarkjs
Proof Size	~200 bytes (3 field elements)
Verification Time	~10ms on-chain
Security Level	128-bit
Privacy	Biometric data never transmitted

4. V6 Economic Model

4.1 Proof of Alive

Proof of Alive solves one of the most fundamental problems in cryptocurrency: what happens to money when people die or disappear? In Bitcoin, an estimated 3-4 million BTC are permanently lost. In Aequitas, AEQ that belongs to inactive wallets eventually returns to the community through a carefully designed escrow system.

Timeline	Action
Year 0–2	Normal usage expected
Year 2	Warning 1 sent on-chain — Guardian can respond
Year 2 + 60 days	Warning 2 sent — Guardian can respond
Year 2 + 120 days	Warning 3 sent — Guardian can respond
Year 2 + 180 days	AEQ moved to PERSONAL ESCROW (not UBI yet)
Year 4	If still inactive → AEQ enters UBI Pool → distributed equally

4.2 Guardian System

The Guardian System protects vulnerable humans — those who are imprisoned, hospitalized, or otherwise unable to access their devices. Every verified human can appoint one Guardian (another verified human) who can confirm they are still alive, with strictly zero transaction rights.

Rule	Detail
1 Guardian per human	Must be another verified human
Only action allowed	confirmAlive() — zero transaction rights
Cannot do	Move funds, transfer AEQ, change any settings
Max wards per Guardian	3 (prevents abuse)
Assignment timelock	7 days (prevents forced assignment under duress)
Circular relationships	Prohibited (A guards B → B cannot guard A)
Guardian chain	Guardian cannot have their own Guardian

4.3 Demurrage — Anti-Hoarding Mechanism

Demurrage is a 1% annual fee on any balance above the holder's fairShare. The money flows to the UBI Pool — it is never deleted. Historical precedent: The town of Wörgl, Austria (1932) introduced

demurrage currency during the Great Depression. Within one year, unemployment dropped 25% while the rest of Austria suffered. The Central Bank shut it down — not because it failed, but because it worked too well and threatened central banking authority.

**Example: fairShare = 1,000 AEQ | Balance = 3,000 AEQ |
Excess = 2,000 AEQ | Monthly fee = 1.67 AEQ → UBI Pool**

4.4 Wealth Cap

Phase	Human Count	Wealth Cap
Phase 0	1 – 100	50× fairShare
Phase 1	101 – 1,000	20× fairShare
Phase 2	1,001 – 10,000	10× fairShare
Phase 3	10,001 – 100,000	5× fairShare
Phase 4	100,000+	3× fairShare

The Wealth Cap is always active from human #1. When any wallet exceeds the cap, the excess is instantly redistributed equally to ALL active humans. For comparison: Bitcoin's top 1% controls 90%+ of supply. In Aequitas, this is mathematically impossible.

4.5 Universal Basic Income

UBI Source	Amount
Transaction fees (0.1%)	20% → UBI Pool
Wealth cap overflow	100% → instant equal redistribution
Demurrage (1% annual on excess)	100% → UBI Pool
Inactive wallet escrow (after 4 years)	100% → UBI Pool

Monthly distribution: $\text{UBI Pool} \div \text{total active humans} = \text{equal payment to everyone}$. This UBI requires no taxation, no government, no political decision. It is a mathematical consequence of the protocol's fairness mechanisms.

5. Registration Flow

Step	Action	Where
1	Fingerprint scan via Hardware Secure Element	Android Device
2	Biometric hash derived (never leaves device)	Android Device
3	Check if already registered (/check endpoint)	Proof Server
4	Groth16 ZKP generated from biometric hash	Proof Server
5	Proof stored, short Proof ID returned	Proof Server
6	MetaMask opens with ?proofId=xxx	Android → Browser
7	Explorer loads proof, auto-connects wallet	Browser/MetaMask
8	User clicks Register On-Chain	Explorer UI
9	Node verifies ZKP, calls AequitasV6 contract	Aequitas Chain
10	V6 state mirrored to PostgreSQL	Node → DB
11	App polls until confirmed	Android App
12	1,000 AEQ credited to wallet	Aequitas Chain

6. Network Infrastructure

Component	URL / Address	Role
Node 1 (Primary)	aequitas-production-9fba.up.railway.app	Primary API, block producer, P2P bootstrap
Node 2 (Secondary)	aequitas-node-2.onrender.com	Secondary API, block producer, P2P peer
Proof Server	aequitas-proof-server-production.up.railway.app	ZKP generation, sybil protection, proof storage
PostgreSQL	Railway Managed Database	Persistent state shared across all nodes
Bootstrap Peer	/dns4/thomas.proxy.rlwy.net/tcp/47298/p2p/...	P2P network entry point

7. Comparison with Existing Systems

Property	Bitcoin	Ethereum	Worldcoin	Aequitas
Supply Mechanism	Mining	Staking	Fixed Issuance	Human Registration
Initial Distribution	Early miners	Founders/VCs	Operators	Equal — all humans
Gini Coefficient	>0.85	>0.70	Unknown	0.00 at genesis
Sybil Protection	None	None	Iris scan	Fingerprint ZKP
Privacy	Pseudonymous	Pseudonymous	Iris stored	Zero biometric data
Wealth Cap	None	None	None	Phase-based (3–50x)
UBI	No	No	Yes (token)	Yes (protocol-native)
Gasless Reg.	No	No	Yes	Yes
Human-only	No	No	Yes	Yes

8. Roadmap

Phase	Status	Milestones
Phase 0	COMPLETE	Smart contracts, ZKP circuit, Android app, Proof Server, BlockDAG chain
V6	LIVE	Sovereign chain (Chain ID 1926), EVM engine, Proof of Alive, Guardian System, Demurrage
Phase 1	IN PROGRESS	APK release, grant applications, growing human registry, Ethereum Foundation application
Phase 2	PLANNED	iOS app, Guardian system activation, Proof of Alive activation, 1,000+ humans
Phase 3	PLANNED	DEX integration, lending protocol, 10,000+ humans, redistribution cycles active
Phase 4	FUTURE	Full decentralization, community governance, cross-chain bridges, 1M+ humans

9. Conclusion

Aequitas represents a fundamental rethinking of what money is and who it should serve. By tying money supply directly to verified human existence, Aequitas creates the first monetary system where equality is not an aspiration but a mathematical property of the system itself.

The V6 protocol addresses the long-term sustainability of this equality through Proof of Alive, the Guardian System, Demurrage, the Wealth Cap, and protocol-native UBI. Each mechanism is designed to preserve equality over time without requiring human intervention, political decisions, or trust in any central authority.

The technology to make this possible has existed for years: Zero-Knowledge Proofs, Hardware Secure Elements, blockchain, smartphones. Aequitas is the first system to assemble these components into a coherent, fair monetary system designed for all of humanity.

"Money exists because people exist. Nothing more, nothing less."

We invite every human being on Earth to register and join a monetary system designed from first principles to be fair to everyone.